Prior-Anchored Deep Learning for Dense UAV Air-to-Ground Channel Knowledge Maps

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***Abstract*—Channel knowledge maps (CKMs) promise fast environment-aware radio planning, but dense air-to-ground CKM generation remains difficult when path loss, delay spread, and angular spread must all be predicted over unseen urban layouts. This paper studies a strict city-holdout evaluation protocol, train-only calibrated physical/statistical priors, and a bounded residual neural model with a Gaussian-mixture residual head. The final model predicts 513×513 dense maps from topology, geometric masks, UAV height, and frozen priors. It is trained through a probabilistic residual/GMM stage followed by deterministic RMSE-oriented fine-tuning; reported maps use the deterministic mixture-mean point prediction. On held-out cities it reaches 1.6519 dB path-loss RMSE, 26.5570 ns delay-spread RMSE, and 11.3854° angular-spread RMSE, reducing RMSE relative to the frozen priors for all three targets. The result supports a hybrid design principle: use calibrated priors for stable radial and visibility-dependent structure, and reserve neural capacity for bounded local residuals and target-distribution mismatch.**

***Index Terms*—channel knowledge maps, UAV communications, radio-map prediction, path loss, delay spread, angular spread, Gaussian mixture models, physics-informed learning.**

I. Introduction

Channel knowledge maps (CKMs) store or predict radio-channel quantities as functions of position and environment, enabling faster planning and adaptation than repeated ray tracing or exhaustive measurements [1], [2]. In UAV-enabled 6G networks, CKMs are especially attractive because altitude changes both the geometry and the line-of-sight probability of the link [3], [4]. A practical CKM surrogate, however, must generalize to urban layouts not seen during training and must not hide hard propagation regimes behind a single global score.

A pure dense image-to-image formulation was not sufficient for this setting. Early U-Net and cGAN models could draw plausible maps, yet their physical errors remained large under a ground-only metric. PMNet-inspired and U-Net-style backbones improved smooth LoS structure but did not solve NLoS mode collapse or the spike-plus-tail nature of spread targets. The final design therefore moved away from "bigger backbone first" and toward a hybrid sequence:

1) define the evaluation contract before comparing models;
2) diagnose target distributions before choosing losses and heads;
3) calibrate strong train-only priors for predictable structure;
4) train a bounded residual model around those frozen priors.

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The contribution is a defensible dense CKM recipe for one fixed UAV simulation contract: strict city holdout, valid-receiver masking, explicit LoS/NLoS and height reporting, and a final prior-anchored neural correction.

II. Related Work

Dense radio-map prediction has often been treated as image-to-image regression. RadioUNet [5] established convolutional radio-map estimation as a strong baseline; PMNet and the ICASSP path-loss challenge family further showed that U-Net-like architectures can perform well for path-loss maps [6]–[8]. More recent models introduce attention, equivariance, or domain-shift machinery, including RMTransformer [9], Radio-GUNet [10], PathFinder [11], Gao et al.'s corridor-weighted multi-layer segmentation model [12], and digital-twin radio-map generators such as AIRMap [13]. These papers provide the closest dense path-loss map context, but they generally do not match the present contract: continuous UAV height, strict city holdout on CKM samples, and three dense targets (path loss, delay spread, angular spread).

CKM work itself emphasizes that a radio map is not merely an image but a position-indexed channel side-information object [1], [2], [14]. This is why the evaluation protocol matters as much as the neural architecture. A random image split can place visually similar samples from the same urban layouts in both training and testing, whereas a city split tests whether the learned propagation surrogate transfers to unseen layouts.

Physics-informed and hybrid methods address a different weakness of pure image regression: the model should not need to rediscover stable propagation effects from pixels alone. ReVeal [15] uses physics-informed constraints for radio-environment mapping, while classical air-to-ground, COST/Hata, and two-ray models encode known distance, height, reflection, and elevation-angle structure [3], [16]–[21]. This paper uses those models only as shape priors; all deployment coefficients are calibrated from training cities under the same CKM masking convention used at inference.

The final neural head is also informed by distribution-aware wireless learning. Heteroscedastic regression [22], mixture-density wireless predictors [23]–[25], and recent stochastic radio-map models [26] all point to the same issue: NLoS and spread targets are not well represented by a single symmetric residual. For spread targets, the closest external references are often scalar channel-parameter predictors or standard channel-model statistics [20], [27]–[29], not dense maps, so they are not used as direct CKM benchmarks here.

TABLE I
EXPERIMENTAL CONTRACT USED FOR ALL REPORTED FINAL NUMBERS.

Item	Setting
Map resolution	513×513, $1\,\mathrm{m} \times 1\,\mathrm{m}$ pixels
Carrier / receiver	7.125 GHz; $h_{\mathrm{rx}} = 1.5\,\mathrm{m}$
Transmitter	One centred UAV; stored $h_{\mathrm{tx}} \approx 12\,\mathrm{m}$ to $478\,\mathrm{m}$, nominal beta law $10\,\mathrm{m}$ to $500\,\mathrm{m}$
Inputs	Topology, ground/visibility masks, frozen priors, scalar height code
Targets	PL (dB), DS (ns), AS (deg), one dense map per target
Metric mask	Non-building ground receivers only; building pixels excluded
Split policy	City holdout; validation selects the checkpoint, test is final only
Samples	Train 10840; validation 2750; test 2590; all 16180

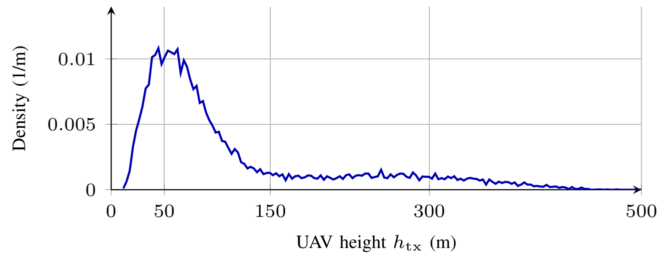

Fig. 1. Empirical UAV transmitter-height density read directly from the CKM HDF5 database using the 3 m histogram of Eq. 2, linearly interpolated between bin centres for display. Most training support lies in the low and mid-altitude bands.

III. Task and Evaluation Protocol

Each sample is a 513×513 urban topology map from the ray-traced CKM setting used in this work. The carrier is fixed at 7.125 GHz, so frequency is not a learned variable; the main transmitter degree of freedom is height. There is one UAV transmitter at the map centre. The transmitter horizontal position is fixed, but its antenna height h_{tx} varies by sample and is treated as a central conditioning variable rather than as metadata. The spatial resolution is $1\,\mathrm{m} \times 1\,\mathrm{m}$ per pixel. Every non-building pixel is treated as an active receiver location, with the receiver antenna placed at $h_{\mathrm{rx}} = 1.5\,\mathrm{m}$ above the ground. Building pixels are excluded from the receiver set rather than treated as easy zero-error targets. The model predicts three aligned dense maps:

$$y_t \in \{\mathrm{PL\ (dB)}, \mathrm{DS\ (ns)}, \mathrm{AS\ (deg)}\}.$$

Let $m_i(p) = 1$ if pixel p of sample i is one of these valid ground receivers, and zero otherwise. For a split $\mathcal{S}$, the target-specific pixel-weighted RMSE is

$$\mathrm{RMSE}_t = \sqrt{\frac{\sum_{i \in \mathcal{S}} \sum_p m_i(p) \left(\hat{y}_{i,t}(p) - y_{i,t}(p)\right)^2}{\sum_{i \in \mathcal{S}} \sum_p m_i(p)}}. \quad (1)$$

The split is city-based rather than random over pixels or tiles. This matters: random splits would allow the model to see nearly identical urban morphology during training and testing. The final held-out test split contains 2590 samples from cities not used to optimize model weights. Validation is used for model selection; test is used only for the final generalization claim. Table II lists the exact city assignment.

The transmitter-height distribution is intentionally non-uniform. Direct comparison against the stored HDF5 histogram shows that the closest tested affine version of the approximate double-beta sampler is the original $h_{\mathrm{design}} \approx 10 + 490X$, with

$$X \sim 0.75\,\mathrm{Beta}(3, 23) + 0.25\,\mathrm{Beta}(4, 4),$$

and nominal support $10\,\mathrm{m}$ to $500\,\mathrm{m}$. The results below use the empirical HDF5 realization rather than the ideal continuous law. Across the 16180 samples, the stored h_{tx} values span $11.61\,\mathrm{m}$ to $477.54\,\mathrm{m}$, with mean $114.02\,\mathrm{m}$ and median $75.66\,\mathrm{m}$. Under the same $10+490X$ law, heights above $478\,\mathrm{m}$ are possible but very rare: $P(h_{\mathrm{tx}} > 478\,\mathrm{m}) = 3.19 \times 10^{-5}$ per draw, or only 0.52 expected samples in $N = 16180$. Drawing no samples above $478\,\mathrm{m}$ has probability about 0.60, making it a statistically normal finite-sampling outcome. The HDF5 maximum $477.54\,\mathrm{m}$ is therefore consistent with the sampler rather than evidence of truncation. Let $\Delta h = 3\,\mathrm{m}$, $B_k = [10+3k,\ 13+3k)$, and h_i be the stored height of sample i. The actual distribution used for reporting is the empirical three-meter histogram

$$\hat{f}_{\Delta h}(h) = d_k, \quad h \in B_k,$$
$$d_k = \frac{n_k}{N \Delta h}, \quad n_k = \sum_{i=1}^{N} \mathbf{1}\{h_i \in B_k\}, \quad N = 16180. \quad (2)$$

For readability in Fig. 1, the plotted curve linearly interpolates between the bin-centre values (c_k, d_k), with $c_k = 11.5 + 3k$. The all-database height counts are 3972 samples below $50\,\mathrm{m}$, 8456 from $50\,\mathrm{m}$ to $150\,\mathrm{m}$, 2490 from $150\,\mathrm{m}$ to $300\,\mathrm{m}$, and 1262 above $300\,\mathrm{m}$; the corresponding training counts are 2676, 5654, 1644, and 866 samples.

A. Quantized Targets and Continuous Outputs

The released CKM targets are not ideal real-valued measurements. Path loss is stored on an approximately 1 dB grid, while delay spread and angular spread are strongly integer-valued and visibly staircase-like in many maps. The model outputs are not quantized to these grids: the residual network emits continuous values, and all reported metrics compare those continuous outputs against the stored targets. This matters for interpretation. The final path-loss MAE is 0.9298 dB, close to the 1 dB label step, while the path-loss RMSE is 1.6519 dB because RMSE emphasizes the remaining hard tails. A cleaner or less quantized dataset would likely expose more of the continuous surrogate's resolution, especially for small residual improvements and spread-map boundaries.

IV. Method

A. Why the Final Method Is Hybrid

The development path exposed three distinct target regimes. First, LoS path loss contains stable radial, height-dependent, and coherent two-ray structure. Second, NLoS path loss is not just a smooth residual; it is a blocked-link distribution-placement problem. Third, delay spread and angular spread contain near-zero LoS spikes and NLoS-heavy tails. Thus a

TABLE II
CITY-HOLDOUT ASSIGNMENT USED BY THE DISTRIBUTION-FIRST AND FINAL NEURAL MODELS.

Split	Samples	Cities	City names
Train	10840	37	Abidjan, Abu Dhabi, Alexandria, Antigua Guatemala, Athens, Bangalore, Bangkok, Bath, Berlin, Bratislava, Brugge, Buenos Aires, Canberra, Chefchaouen, Dalian, Hanoi, Ho Chi Minh City, Hyderabad, Kinshasa, Leuven, Ljubljana, Luanda, Lyon, Manaus, Minsk, Nagpur, Paris, Rio de Janeiro, Santiago, Shanghai, Siem Reap, St. Louis, Thane, Toledo, Valparaiso, Venice, Vienna.
Validation	2750	15	Bilbao, Chiang Mai, Cleveland, Dhaka, Dubrovnik, Fortaleza, Marrakesh, Mexico City, Munich, Nazareth, Nice, Salvador, Segovia, Tunis, Victoria.
Test	2590	14	Barcelona, Beijing, Carcassonne, Copenhagen, Halifax, Jaipur, Johannesburg, Key West, Kuala Lumpur, Osaka, Phuket, Pune, Surat, Vancouver.

TABLE III
DESIGN PATH BEHIND THE FINAL MODEL. HISTORICAL DIAGNOSTICS ARE INCLUDED TO EXPLAIN THE CHOICE OF HEAD AND PRIOR, NOT AS THE FINAL LEADERBOARD.

Stage	Role	Main lesson
Direct dense CNNs / cGANs	Early baselines	Could draw plausible CKM texture, but did not reliably solve NLoS tails or spread quantization under the ground-only city-holdout metric.
Earlier no-prior GMM heads	Distribution diagnostic	Worse than the final system, but showed that PL, DS, and AS are jointly predictable when the head represents LoS/NLoS and heavy-tailed targets.
Frozen PL/DS/AS priors	Structured anchor	Captured most radial, visibility, height, and spread-scale structure using train-only calibration.
Final prior-anchored residual GMM	Reported model	Keeps the distribution-aware head, but moves it into bounded residual space around the frozen priors and then fine-tunes the deterministic point map.

single point-regression CNN is asked to solve three different statistical problems at once.

The final method uses a division of labour. Frozen priors provide a strong mean prediction for structure that can be calibrated reliably from training data. The neural model receives those priors and learns bounded residual corrections. This makes the prior useful but not magical: distribution-first neural models showed that NLoS structure can be learned without explicit priors when the target distribution is represented correctly. The priors are reintroduced because they are cheaper, more stable, and more auditable for the predictable part of the map.

In this paper, a *prior* means a frozen estimator fitted only on training cities, not a hand-tuned curve evaluated on the test set. The prior sees the same support information available at deployment: centred transmitter geometry, UAV height, topology-derived morphology, and the geometric LoS/NLoS mask. Once its coefficients and fallback tables are stored, no gradient update is applied to it inside the neural training loop. This separation is important because it lets the final residual model be judged against a demanding baseline: the model must improve an already calibrated predictor rather than merely beat FSPL.

B. Frozen Priors

The path-loss prior combines a coherent two-ray LoS branch with a calibrated NLoS branch. More precisely, Eq. (4) is the Friis/free-space dB term and Eq. (5) is the standard coherent direct plus ground-reflected image-source construction [16], [17], also used in low-altitude A2G two-ray modeling [21]. The fitted ρ, ϕ, b, and r curves are not taken from those references; they are CKM train-split calibration terms wrapped around that two-ray structure. The NLoS branch uses the COST231/Hata urban-macro expression as a coarse blocked link trend [18], combines it with an Al-Hourani-style elevation-dependent A2G excess-loss envelope [19], and keeps the LoS/NLoS regime separation used in UAV and standardized channel-model literature [3], [20]. The spread priors use the log-domain treatment of large-scale delay/angle spread parameters from 3GPP/WINNER-style models [20], [27]; the UAV empirical literature motivates the elevation and angular-spread terms rather than supplying the CKM coefficients directly [28], [29]. These references define the shape assumptions; all numerical coefficients used here are still fitted on the training cities only. The UAV remains horizontally centred, so d_{2D} is the per-pixel distance from the centre transmitter to each receiver. The height $h_{\rm tx}$, however, changes from sample to sample and is one of the main physical degrees of freedom in the method. For receiver height $h_{\rm rx}$ and carrier frequency $f_{\rm MHz}$, the direct and reflected distances are

$$\begin{aligned} d_{\rm LoS} &= \sqrt{d_{2D}^2 + (h_{\rm tx} - h_{\rm rx})^2}, \\ d_{\rm ref} &= \sqrt{d_{2D}^2 + (h_{\rm tx} + h_{\rm rx})^2}. \end{aligned} \tag{3}$$

For the positive UAV and receiver heights used here, $d_{\rm ref}$ is strictly larger than $d_{\rm LoS}$; equality would occur only in the limiting geometric case where one antenna lies exactly on the ground plane. The LoS branch starts from

$$\mathrm{FSPL} = 32.45 + 20\log_{10}\left(\frac{d_{\rm LoS}}{1000}\right) + 20\log_{10}(f_{\rm MHz}) \tag{4}$$

and adds an effective coherent reflection correction

$$C_{\rm 2ray} = -20\log_{10}\left|1 + \rho(h_{\rm tx})\frac{d_{\rm LoS}}{d_{\rm ref}} e^{-j(k(d_{\rm ref} - d_{\rm LoS}) + \phi(h_{\rm tx}))}\right|. \tag{5}$$

The final LoS prior is

$$\mathrm{PL}_{\rm LoS} = \mathrm{FSPL} + C_{\rm 2ray} + b(h_{\rm tx}) + r(d_{2D}, h_{\rm tx}), \tag{6}$$

where b is a height-dependent bias and r is a smoothed radial residual fit. This LoS prior is not FSPL alone and it is not a learned linear regressor. The direct distance $d_{\rm LoS}$ sets the free-space baseline, the image-source distance $d_{\rm ref}$ supplies the ground-reflected path, and $k(d_{\rm ref} - d_{\rm LoS})$ is the extra phase accumulated by that reflected wave, with $k = 2\pi f/c$ the free-space wavenumber. The train-calibrated functions $\rho(h_{\rm tx})$ and $\phi(h_{\rm tx})$ are effective reflection amplitude and phase curves, fitted in 5 m UAV-height bins, smoothed across bins, and linearly interpolated to the actual sample height at inference. The

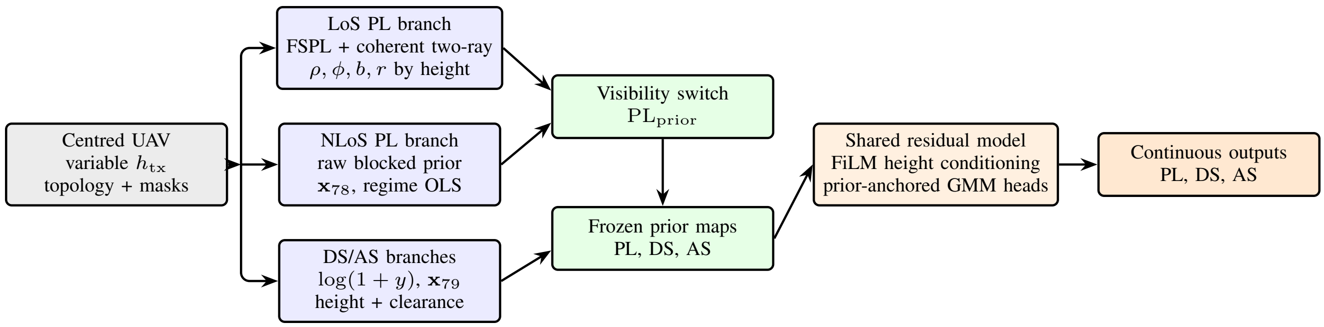

Fig. 2. Compact prior-anchored pipeline. The transmitter is always centred, but its height varies by sample. Frozen height-aware priors are produced first and then used as channels for the final prior-anchored residual GMM model.

terms $b(h_{\mathrm{tx}})$ and $r(d_{2D}, h_{\mathrm{tx}})$ are outside the textbook two-ray expression: they remove, respectively, a remaining height-dependent offset and a smooth range-dependent residual; these terms are also stored as height-conditioned calibration curves rather than as one global correction. The deployed LoS path-loss prior is finally kept inside the $[20, 180]$ dB range used by the final model. For this CKM dataset, valid calibrated LoS values are not expected to leave that interval, so the operation is a defensive range guard rather than a correction that should materially change the prior. The notation in this compact paper is local: $\rho(h_{\mathrm{tx}})$ is the LoS reflection-amplitude curve, whereas ρ_k denotes a local density fraction; likewise $b(h_{\mathrm{tx}})$ is a LoS dB bias, while b_q^{nat}, b_{top}, and b_{41} denote spread anchors, spread topology offsets, and blocker-depth features. These symbols should not be read as one shared physical variable.

The LoS calibration is deliberately structured rather than fully flexible. It is fitted only on valid LoS ground pixels from training cities. First, the effective reflection amplitude ρ and phase ϕ are fitted in 5 m UAV-height bins to reproduce the ring-like constructive and destructive interference visible in LoS path-loss maps. Second, the per-height bias b removes a remaining vertical offset after the coherent two-ray fit. Third, the radial residual table $r(d_{2D}, h_{\mathrm{tx}})$ averages the remaining error by distance from the centred transmitter, smooths it across radius, clips it to a small dB range, and stores it by height. At inference, the actual h_{tx} selects/interpolates these calibration curves, so the LoS prior changes continuously with antenna height even though the UAV horizontal coordinate remains fixed at the map centre.

The reason for keeping this branch so explicit is diagnostic. In the CKM maps, many LoS errors appear as rings around the centred transmitter rather than as building-edge artifacts. A generic CNN can approximate those rings, but it must spend capacity on a mostly one-dimensional height/radius phenomenon. The two-ray-plus-radial table removes that component before the residual model sees the sample, leaving the network to focus on local corrections near visibility transitions and morphology-dependent errors.

The NLoS prior starts from the maximum of COST231-like and A2G-NLoS raw terms. The first term below is the COST231/Hata urban-macro formula adapted as a shape prior for the CKM geometry, not as an assertion that the original model is directly valid at every UAV height:

$$
\begin{aligned}
\mathrm{PL}_C &= 46.3 + 33.9 \log_{10}(f_{\mathrm{MHz}}) \\
&\quad - 13.82 \log_{10}(h_{\mathrm{tx}}) - a(h_{\mathrm{rx}}) \\
&\quad + \eta(h_{\mathrm{tx}}) \log_{10}(d_{\mathrm{km}}) + 3, \\
\eta(h_{\mathrm{tx}}) &= 44.9 - 6.55 \log_{10}(h_{\mathrm{tx}}),
\end{aligned}
\tag{7}
$$

where C denotes the COST231-shaped branch, $d_{\mathrm{km}} = \max(d_{2D}/1000, 0.001)$, and $a(h_{\mathrm{rx}}) = (1.1 \log_{10} f_{\mathrm{MHz}} - 0.7)h_{\mathrm{rx}} - (1.56 \log_{10} f_{\mathrm{MHz}} - 0.8)$ is the small/medium-city mobile-height correction. The $+3$ dB term is retained as the COST231 metropolitan correction inside this empirical basis; it is not a claim that the original COST231 deployment domain transfers unchanged to 7.125 GHz UAV links. The second term is an A2G elevation-dependent NLoS envelope: it keeps the altitude/elevation structure of coverage-oriented A2G models, with fixed excess-loss constants later corrected by the train-only NLoS calibration. With the positive fitted envelope constant used here, this raw term is deliberately conservative and should not be read as a standalone monotonic claim that every higher-elevation NLoS link has lower loss:

$$
\begin{aligned}
\mathrm{PL}_{\mathrm{A2G,NLoS}} &= \lambda_0(h_{\mathrm{tx}}, f) + \beta_{\mathrm{NLoS}} \\
&\quad + A_{\mathrm{NLoS}} \exp\left(-\frac{90 - \theta^\circ}{\tau_{\mathrm{NLoS}}}\right),
\end{aligned}
\tag{8}
$$

with $\lambda_0 = 20 \log_{10}(4\pi h_{\mathrm{tx}} f/c)$, $\beta_{\mathrm{NLoS}} = -16.16$ dB, $A_{\mathrm{NLoS}} = 12.0436$, and $\tau_{\mathrm{NLoS}} = 7.52^\circ$. The raw blocked-link prior is the conservative envelope

$$
P_0 = \mathrm{PL}_{\mathrm{NLoS,raw}} = \max(\mathrm{PL}_C, \mathrm{PL}_{\mathrm{A2G,NLoS}}). \tag{9}
$$

This maximum is deliberately pessimistic: if the urban empirical curve predicts stronger attenuation it is kept, and if the elevation-aware A2G envelope is larger it is kept instead. The raw prior is then corrected by train-only regime-wise linear calibration:

$$
\mathrm{PL}_{\mathrm{NLoS}} = \mathrm{clip}(\mathbf{x}_{78}^\top \mathbf{w}_{78,r}, 20, 180). \tag{10}
$$

The path-loss prior is therefore not one formula with a small correction. It is two different calibrated feature extractors. The LoS side calibrates height-binned two-ray amplitude/phase, height bias, and radial residual curves. The NLoS side adds height-, building-topology-, and density-dependent calibration, using the raw blocked-link prior as only one feature in a 15-term morphology vector:

$$\begin{aligned}\mathbf{x}_{78} = [&P_0^2, P_0, \log(1+d_{2D}), \rho_{15}, \rho_{41},\\ &h_{15}, h_{41}, \rho_{41}\log(1+d_{2D}), n_{15},\\ &n_{41}, n_{41}\log(1+d_{2D}), \sigma_{\text{shadow}}, \theta_{\text{norm}},\\ &n_{41}\theta_{\text{norm}}, 1]^\top,\end{aligned} \tag{11}$$

where P_0 is the raw NLoS prior, ρ_{15} and ρ_{41} are the fractions of building pixels in 15×15 and 41×41 windows, h_{15} and h_{41} are local building-height summaries, n_{15} and n_{41} summarize local NLoS support, σ_{shadow} is an elevation-dependent shadowing proxy, and θ_{norm} is the normalized elevation-angle feature. The interaction $\rho_{41}\log(1+d_{2D})$ lets density matter differently with range, $n_{41}\log(1+d_{2D})$ lets local blockage support vary with range, and $n_{41}\theta_{\text{norm}}$ activates when obstruction and elevation jointly matter. The constant 1 is the regime-wise bias term. The deployed path-loss prior is selected by the geometric visibility mask:

$$\text{PL}_{\text{prior}} = m_{\text{LoS}}\text{PL}_{\text{LoS}} + (1-m_{\text{LoS}})\text{PL}_{\text{NLoS}}. \tag{12}$$

The regime index r is not a learned latent class. It is selected from observable scene descriptors: topology/morphology class, LoS/NLoS state, and antenna-height group, with fallback calibrations when an exact regime has too few training pixels. Thus the prior is auditable: one can inspect the raw blocked estimate P_0, the local morphology terms, the selected regime, and the final linear correction separately.

This NLoS branch is intentionally less physical than the LoS branch. The visibility mask says that the direct ray is blocked, but it does not say which street canyon, rooftop edge, or local clutter feature dominates the strongest remaining path. The calibrated feature vector therefore treats COST/A2G values as one input among morphology summaries. It is a compromise between a fully analytic blocked-link law, which was too rigid, and a deep model, which was less transparent when trained from scratch.

Table IV makes this provenance explicit. The only pieces taken directly as literature structures are the Friis/two-ray geometry, the COST231/Hata urban trend, the A2G elevation-loss shape, and the log-domain spread convention. The reflection curves, radial residuals, OLS/ridge weights, fallback hierarchy, and morphology windows are fitted or computed inside this CKM dataset pipeline.

The LoS two-ray parameters are fitted by a small grid search rather than by a neural optimizer. For each 5 m height bin B_h, candidate values $\hat{\rho} \in \{0, 0.05, \ldots, 1.5\}$ and $\hat{\phi} \in [-\pi, \pi)$ on a 48-point phase grid are evaluated on valid LoS training pixels. The upper end of this grid is a robustness allowance for an effective CKM-fitted coefficient, not a claim that a passive Fresnel reflection coefficient exceeds unity; the smoothed deployed reflection-amplitude curve stored by the final prior remains below unity. For each candidate pair, the dB bias has the closed-form value

$$\hat{b}(\hat{\rho}, \hat{\phi}) = \frac{1}{|B_h|}\sum_{i \in B_h}\left[y_i - \text{FSPL}_i - C_{2\text{ray},i}(\hat{\rho}, \hat{\phi})\right], \tag{13}$$

and the selected pair minimizes the corresponding RMSE. The fitted ρ, ϕ, and b curves are then smoothed across neighbouring height bins, weighted by bin support, and linearly interpolated at inference.

The NLoS branch uses the same train-only principle but with a linear morphology calibrator. For every supported regime r, the calibration weights solve

$$\hat{\mathbf{w}}_{78,r} = \arg\min_{\mathbf{w}} \sum_{i \in r}\left(\mathbf{x}_{78,i}^\top \mathbf{w} - y_i\right)^2, \tag{14}$$

with fallback to stored antenna-height keys within the same morphology and LoS/NLoS group when the exact antenna-height key is unavailable. At deployment, the same deterministic features select one stored coefficient vector and produce the clipped NLoS prior in Eq. (10). This is an ordinary least-squares objective, not a requirement that every regime have a nonsingular normal matrix; if a calibration block is rank deficient or ill-conditioned, the stored solution is interpreted as the numerical least-squares or pseudo-inverse solution associated with this objective.

The delay-spread and angular-spread priors are fitted in a transformed domain,

$$z = \log(1+y), \qquad \hat{y} = \exp(\hat{z}) - 1, \tag{15}$$

because both standard channel models and the CKM histograms make spread quantities more regular in log space [20], [27]. The released spread maps have a near-zero mass plus a long tail: LoS direct-path pixels cluster near zero, while blocked or rooftop-scattered pixels create rare but important large values. The standardized models justify the log-domain large-scale variable; the particular range, elevation, density, height, and NLoS-support terms below are CKM-specific deterministic covariates chosen to make that log variable depend on observable map geometry. Their raw prior is

$$\begin{aligned}z^{(0)} = \; &\log(1+b_q^{\text{nat}}) + b_{\text{top}} + a_1\log(1+d_{2D}) + a_2\theta_{\text{inv}}\\ &+ a_3\rho_{41} + a_4 h_{41} + a_5 n_{41} + a_6 n_{41}\theta_{\text{inv}},\end{aligned} \tag{16}$$

The constants in Eq. (16) are fixed before ridge calibration. For delay spread, $b_{\text{LoS}}^{\text{nat}} = 4.0\,\text{ns}$, $b_{\text{NLoS}}^{\text{nat}} = 11.0\,\text{ns}$, and $(a_1, \ldots, a_6) = (0.11, 0.62, 0.48, 0.32, 0.86, 0.60)$. For angular spread, $b_{\text{LoS}}^{\text{nat}} = 3.0^\circ$, $b_{\text{NLoS}}^{\text{nat}} = 7.0^\circ$, and $(a_1, \ldots, a_6) = (0.05, 0.52, 0.60, 0.38, 0.58, 0.35)$. The topology offsets b_{top} are log-domain offsets, not native ns/degree offsets. In the class order open-sparse low-rise, open-sparse vertical, mixed-compact low-rise, mixed-compact mid-rise, dense-block mid-rise, and dense-block high-rise, the delay-spread offsets are $(0.00, 0.18, 0.06, 0.22, 0.24, 0.34)$, and the angular-spread offsets are $(0.00, 0.09, 0.05, 0.16, 0.18, 0.24)$. The six topology classes are assigned from map-level building coverage $\bar{\rho}_{\text{map}}$ and mean non-ground building height $\bar{h}_{\text{map}}$: coverage $\bar{\rho}_{\text{map}} \leq 0.12$ is open sparse, with low-rise if $\bar{h}_{\text{map}} \leq 12\,\text{m}$ and vertical otherwise; coverage $\bar{\rho}_{\text{map}} \geq 0.28$ is dense block, with mid-rise if $\bar{h}_{\text{map}} \leq 28\,\text{m}$ and high-rise otherwise; intermediate coverage is mixed compact, with low-rise if $\bar{h}_{\text{map}} \leq 12\,\text{m}$ and mid-rise otherwise. These labels are map-level routing keys for calibration, not per-pixel learned features.

Here q is the LoS/NLoS state. The native anchor b_q^{nat} gives a different starting scale to LoS and NLoS pixels, b_{top} shifts the baseline by topology class, $\log(1+d_{2D})$ captures range

TABLE IV
TERM-BY-TERM PROVENANCE OF THE MAIN FROZEN-PRIOR QUANTITIES.

Term	Prior block	Source	Role in this paper
Literature formula structures			
d_{LoS}	LoS PL	CKM geometry + Friis distance term [16], [17]	Direct transmitter–receiver distance used in FSPL.
d_{ref}	LoS PL	Image-source two-ray geometry [16], [17], [21]	Length of the single ground-reflected ray.
$C_{2\mathrm{ray}}$	LoS PL	Coherent direct/reflected field sum [16], [17]	Constructive/destructive radial rings before learned correction.
P_0	NLoS PL	COST231/Hata trend [18] + A2G elevation envelope [3], [19]	Raw blocked-link scale before morphology OLS calibration.
CKM deterministic geometry and morphology features			
ρ_k, h_k, n_k	NLoS PL and spreads	Topology and LoS/NLoS masks	Local density, building height, and NLoS support in a $k \times k$ window.
b_{41}, c_{41}, t_{41}	Spreads	Obstruction features motivated by A2G clutter behavior [3], [28], [29]	Blocker depth, transmitter clearance, and taller-building fraction.
$\theta_{\mathrm{norm}}, \theta_{\mathrm{inv}}$	Spreads	Geometry + A2G elevation dependence [3], [19], [20]	Elevation angle and low-elevation complement for blocked/spread-prone regimes.
CKM train-split fitted calibration			
$\rho(h_{\mathrm{tx}}), \phi(h_{\mathrm{tx}})$	LoS PL	Train-split fit + smoothing	Smoothed deployed effective reflection amplitude and phase; not material constants.
$b(h_{\mathrm{tx}})$	LoS PL	Train-split fit + smoothing	Smoothed deployed height-dependent offset left after coherent two-ray fitting.
$r(d_{2D}, h_{\mathrm{tx}})$	LoS PL	Train-split fit + smoothing	Smoothed deployed radial residual left by two-ray plus bias.
$b_q^{\mathrm{nat}}, b_{\mathrm{top}}, a_j$	Spreads	Fixed CKM raw-prior constants	Initial log-domain DS/AS scale before ridge calibration.

TABLE V
FROZEN-PRIOR ESTIMATORS USED BY THE FINAL MODEL. ALL COEFFICIENTS ARE FITTED ON TRAINING CITIES AND FROZEN BEFORE NEURAL RESIDUAL TRAINING; CITATIONS INDICATE THE LITERATURE BASIS OF EACH STARTING STRUCTURE.

Target/regime	Starting structure	Train-only calibration	Deployed prior map
PL LoS	FSPL plus coherent direct/reflected two-ray correction [16], [17], [21]	$\rho(h_{\mathrm{tx}})$, $\phi(h_{\mathrm{tx}})$, $b(h_{\mathrm{tx}})$, and radial residual $r(d_{2D}, h_{\mathrm{tx}})$ in smoothed 5 m height bins	LoS path-loss map carrying the dominant height/radius rings before any neural residual is added.
PL NLoS	Conservative envelope of COST231-shaped urban loss and an elevation-aware A2G-NLoS term [3], [18]–[20]	Regime-wise OLS over $\mathbf{x}_{78}$, selected by morphology, visibility, and antenna-height group with fallback keys and output clipping	NLoS path-loss map that stores blocked-link scale, local density, blocker, and elevation effects.
DS / AS	Log-domain raw spike-plus-tail scale from visibility, range, elevation, and local morphology [20], [27]–[29]	Ridge calibration over $\mathbf{x}_{79}$ per metric/regime, with height-collapsed, topology-wide, visibility-wide, and global fallbacks	Delay- and angular-spread maps in native units after $\exp(\hat{z}) - 1$, used as frozen statistical anchors.

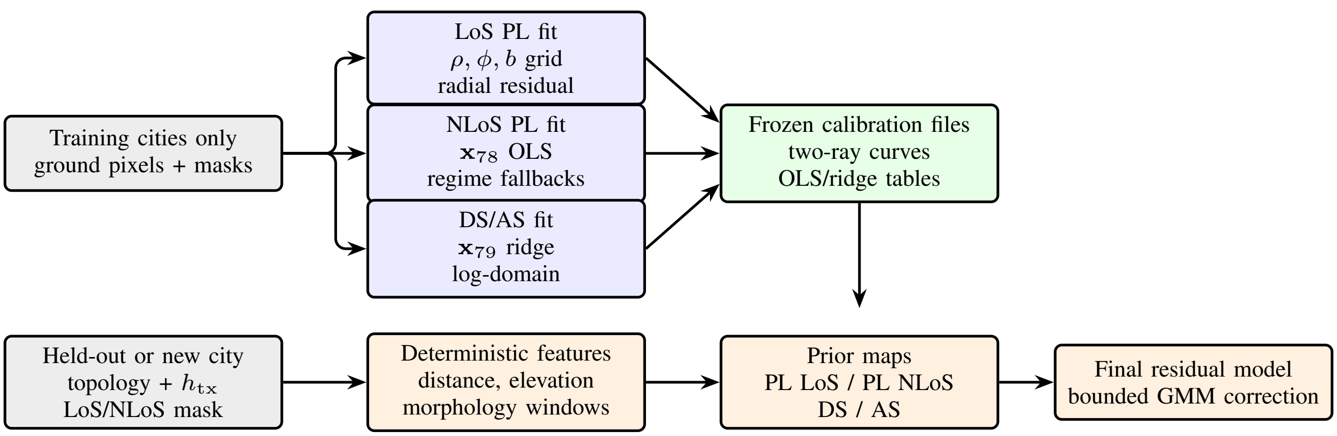

Fig. 3. Train-only prior calibration and deployment. The upper row fits and stores the prior parameters using training cities only. The lower row applies those frozen tables to a held-out sample before the neural residual model is called.

growth, θ_{inv} increases the prior for low-elevation links, and ρ_{41}, h_{41}, and n_{41} summarize local building density, height, and NLoS support. This raw log prior is intentionally simple; it encodes the expected spike-plus-tail shape before CKM-specific calibration. A 23-term feature vector then augments $z^{(0)}$ with range, elevation, height, multi-scale morphology, blocker-depth, clearance, and interaction terms:

$$\begin{aligned}\mathbf{x}_{79} = [&(z^{(0)})^2, z^{(0)}, \log(1+d_{2D}), \theta_{\mathrm{norm}}, \theta_{\mathrm{inv}}, h, h^2,\\ &\rho_{15}, \rho_{41}, h_{15}, h_{41}, n_{15}, n_{41}, n_{41}\log(1+d_{2D}),\\ &\rho_{41}\theta_{\mathrm{inv}}, b_{41}, 1, c_{41}, t_{41}, \theta_{\mathrm{norm}}\rho_{41},\\ &c_{41}\theta_{\mathrm{inv}}, t_{41}\rho_{41}, t_{41}n_{41}]^\top.\end{aligned} \tag{17}$$

Here h is normalized UAV height, b_{41} is a local blocker-depth summary, c_{41} is transmitter clearance over nearby buildings, and t_{41} is the fraction of locally taller buildings. These height terms are not incidental: they tell the spread prior whether the

TABLE VI
FROZEN-PRIOR-ONLY RMSE ON THE FINAL HELD-OUT TEST SPLIT. THESE ARE THE ANCHORS THAT THE RESIDUAL MODEL MUST IMPROVE.

Target	Overall	LoS	NLoS	Unit
Path loss	1.9383	1.7519	3.5143	dB
Delay spread	28.1023	27.4851	29.6157	ns
Angular spread	13.7416	15.2818	8.6614	deg

centred UAV is above the local skyline, near rooftop level, or below nearby high-rise clutter. The quadratic raw prior terms let the calibration bend the initial log-domain estimate, while the interaction terms represent cases where range, low elevation, dense buildings, and NLoS support become harmful together; in particular, $c_{41}\theta_{\rm inv}$ means that transmitter clearance is allowed to matter differently at low elevation, not that clearance is only important when NLoS support is already high. The calibrated log prior is ridge-regularized per metric and regime:

$$\hat{\mathbf{w}}_{79,r} = \left(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{D}^\top\mathbf{D}\right)^{-1}\mathbf{X}^\top\mathbf{z}. \tag{18}$$

The calibrated log prior is then $\hat{z} = \mathbf{x}_{79}^\top\hat{\mathbf{w}}_{79,r}$. The diagonal regularizer $\mathbf{D}$ keeps the calibrated curve close to the raw prior unless the training residuals consistently justify a correction. In practice this matters for the spread targets: a flexible unregularized fit can chase rare high-spread pixels, whereas the ridge fit keeps the near-zero LoS bulk and the NLoS tail connected to the same physical features.

The spread prior should be read as a statistical prior rather than as a ray-by-ray multipath model. It does not attempt to reconstruct individual reflections. Instead, it predicts the expected scale of a dense spread map from height, range, LoS/NLoS state, and local morphology. The log transform makes the near-zero mass and the long tail share one regression space, while the regime fallbacks prevent small calibration subsets from inventing unsupported high-spread behaviour.

All calibrations use training cities only. The spread-prior dictionary contains 114 fitted regime keys, but this number is not one Cartesian product. Per spread metric, the exact grid contributes 6 topology classes $\times$ 2 visibility states $\times$ 3 height bins $=$ 36 exact keys. The implementation then adds 12 height-collapsed topology/visibility fallbacks, 6 topology-wide fallbacks, 2 global visibility fallbacks, and 1 fully global fallback, for 57 keys per metric and 114 keys across DS and AS. Height is handled differently across the frozen estimators: the LoS PL branch interpolates its fitted 5 m height-bin curves, while the NLoS PL and spread branches use discrete antenna-height regimes with fallback keys; the spread model also keeps continuous height, clearance, and taller-building features inside $\mathbf{x}_{79}$. This heavy calibration is the point: the prior stores many small decisions about morphology, visibility, and height before the neural residual is trained. On the final held-out test split, the deployed prior anchors are summarized in Table VI. Beyond the topology, masks, and scalar height path, the final neural model receives the deployed prior maps rather than the handcrafted feature vectors; those features remain inside the frozen prior estimators instead of becoming additional learned inputs.

C. Prior-Anchored Residual Model

The neural model receives nine aligned channels: topology, LoS mask, NLoS mask, ground mask, combined path-loss prior, LoS path-loss prior, NLoS path-loss prior, delay-spread prior, and angular-spread prior. Height is encoded with sinusoidal features and injected through FiLM modulation [30], so the final network can adapt the residual correction continuously between the calibrated height regimes instead of learning one height-agnostic correction. The backbone is a shared U-Net-style encoder-decoder with GroupNorm, FiLM conditioning, and task-specific residual heads.

The actual final model is therefore not a replacement for the priors. Its input tensor already contains the deployed path-loss and spread estimates, and its output head is constrained to make bounded corrections around them. This choice changes the learning problem: the encoder-decoder does not need to infer absolute free-space attenuation, altitude trends, or the average spread scale from pixels alone. It learns where the frozen estimates are locally biased and how the remaining residual distribution should be placed across the map.

The height path mirrors the full implementation. The scalar $h_{\rm tx}$ is expanded to a 32-dimensional sinusoidal code using 16 log-spaced frequencies over the UAV-height range,

$$\mathbf{e}(h_{\rm tx}) = \left[\sin\frac{h_{\rm tx}}{f_i}, \cos\frac{h_{\rm tx}}{f_i}\right]_{i=1}^{16} \in \mathbb{R}^{32}, \tag{19}$$

then mapped by an MLP to a $\mathbb{R}^{160}$ condition vector. Each FiLM site projects this vector to channel-wise scale and shift parameters and modulates a feature map as

$$\mathbf{X}' = \mathbf{X} \odot (1 + \mathbf{g}) + \mathbf{b}. \tag{20}$$

Encoder FiLM uses the height condition directly; decoder FiLM and the global task heads use a fused condition formed from the height code and the pooled bottleneck context. Thus the same centred topology can receive different residual corrections when the UAV is at 30 m, 150 m, or 350 m. The height-aware results in Sec. V show that this is not only an architectural convenience: sinusoidal height embeddings with FiLM modulation are sufficient for one model to cover the full varying-height range without training a separate network per altitude bin.

The head keeps the two-stage logic of the distribution-first predecessors, but moves it into residual space around the frozen prior. The global stage asks: "what residual distribution does this whole map, height, and topology imply?" The local stage asks: "where should each residual component be placed in the 513×513 map?" This is the key distinction from a direct U-Net regressor. The network does not directly overwrite the prior with a free-form map; it predicts a small residual distribution and then assigns that residual pixel by pixel.

For each task k, visibility region $r \in \{\mathrm{LoS}, \mathrm{NLoS}\}$, component $j \in \{1,2,3\}$, and pixel p, the global head receives the pooled bottleneck and fused height/context vector and outputs

$$\{\pi^{(g)}_{k,r,j}, \delta^{(g)}_{k,r,j}, \sigma^{(g)}_{k,r,j}\}_{j=1}^{3}.$$

After the implementation transforms, $\pi^{(g)}$ is a map-level residual mixture distribution, $\delta^{(g)}$ is a bounded global residual

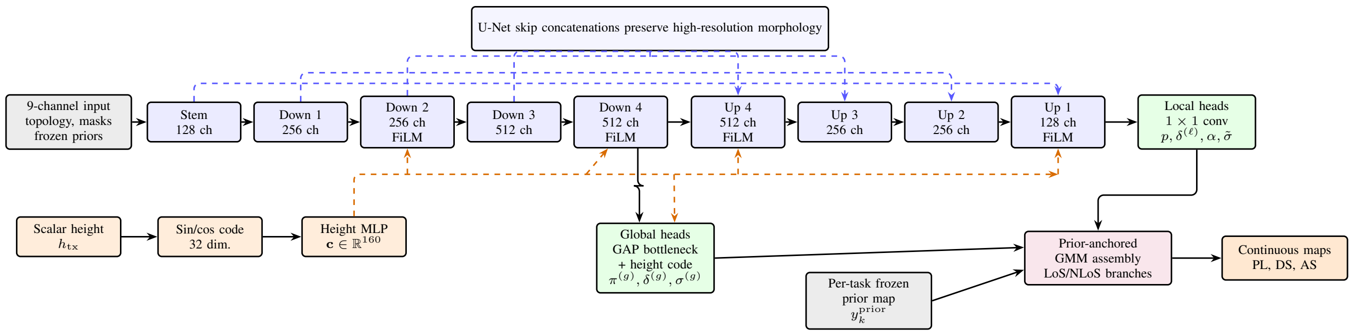

Fig. 4. Final shared deep-learning architecture used in the reported system diagram. The map stack carries topology, masks, and frozen priors; the scalar UAV height is encoded separately and injected through FiLM. Encoder FiLM uses the height code; decoder FiLM and global heads use the fused height/bottleneck context. A shared U-Net produces local residual fields, pooled bottleneck features produce global GMM parameters, and the prior-anchored GMM assembly emits continuous PL/DS/AS maps. The blue dashed arrows show the four encoder–decoder concatenations used in the implementation.

offset, and $\sigma^{(g)}$ is a bounded global uncertainty scale. The local head receives the full-resolution decoder feature map and outputs

$$\{p_{k,r,j}(p)\}_{j=1}^{3}, \quad \delta^{(\ell)}_{k,r}(p), \quad \alpha_{k,r}(p), \quad \tilde{\sigma}_{k,r}(p).$$

Here $p_{k,r,j}(p)$ is the per-pixel component assignment, the local residual $\delta^{(\ell)}$ shifts all components at that pixel, α is an anchor gate initialized with bias -2 so the model starts close to the prior, and $\tilde{\sigma}$ is the local uncertainty term. This extends the distribution-aware line of work from scalar mixture predictors to dense residual maps [23]–[25]. The native residual is clipped by target and visibility:

$$\Delta_{k,r,j}(p) = \operatorname{clip}\left(\delta^{(g)}_{k,r,j} + \delta^{(\ell)}_{k,r}(p), -\delta_{\max,k,r}, \delta_{\max,k,r}\right), \tag{21}$$

with caps of $7/25$ dB for PL LoS/NLoS, $20/20$ ns for DS, and $17^\circ/15^\circ$ for AS. The residual cap is specified in native units: dB for PL, ns for DS, and degrees for AS. Omitting indices for compactness, the component mean is anchored to the frozen prior after this native residual has been assembled:

$$\begin{aligned} u^{\text{nat}} &= y^{\text{prior,nat}} + \alpha\Delta, \\ \mu &= u^{\text{nat}}, \qquad k = \text{PL}, \\ \mu &= \log(1 + [u^{\text{nat}}]_+), \qquad k \in \{\text{DS}, \text{AS}\}. \end{aligned} \tag{22}$$

where $[u^{\text{nat}}]_+ = \max(u^{\text{nat}}, 0)$ and $0 \le \alpha_{k,r}(p) \le 1$. PL is modeled in native dB space, while DS and AS use the log transform in Eq. (15). The deterministic map used for RMSE is the mixture mean in the modelled space:

$$\hat{z}_{k,r}(p) = \sum_{j=1}^{3} p_{k,r,j}(p)\mu_{k,r,j}(p). \tag{23}$$

For PL, $\hat{y}_{k,r} = \hat{z}_{k,r}$. For DS and AS, the output is mapped back with $\hat{y}_{k,r} = \exp(\hat{z}_{k,r}) - 1$. Thus the spread output is the deterministic point map used for RMSE, not the exact native-space expectation of the log-space mixture. The final map selects the LoS or NLoS branch with the explicit geometric mask. The mixture variances combine map-level and local uncertainty,

$$\sigma_{k,r,j}(p) = \sqrt{\left(\sigma^{(g)}_{k,r,j}\right)^2 + (\tilde{\sigma}_{k,r}(p))^2}, \tag{24}$$

with both terms constrained to $[0.05, 3.0]$. During the residual/GMM stage these variances and mixture weights define a pixel-wise likelihood; during the final fine-tuning stage the same head remains in place but the reported output is the deterministic point map above.

The two mixture-weight objects have different jobs. The global weights $\pi^{(g)}_{k,r,j}$ describe the sample-level residual histogram and are used by the distributional regularizer. The local weights $p_{k,r,j}(p)$ are the pixel-wise assignment probabilities used in the final dense map. During the residual/GMM stage the per-pixel density is

$$q_{k,r}(z_k(p) \mid p) = \sum_{j=1}^{3} p_{k,r,j}(p)\,\mathcal{N}\left(z_k(p); \mu_{k,r,j}(p), \sigma^2_{k,r,j}(p)\right), \tag{25}$$

and the regional NLL is the masked average of $-\log q_{k,r}(z_k(p) \mid p)$ over valid LoS or NLoS pixels. Separately, the KL term bins the true residuals $\epsilon_k(p) = y_k(p) - y_k^{\text{prior}}(p)$ into a soft empirical histogram and compares it with the histogram implied by $\{\pi^{(g)}_{k,r,j}, \delta^{(g)}_{k,r,j}\}$. This is what makes the head "distribution-first": before the decoder places values pixel by pixel, the global branch is pressured to put mixture mass where the map residuals actually live.

This head is useful even when the final reported number is a deterministic RMSE. The probabilistic stage teaches the network that a hard NLoS region can contain several plausible residual levels, while the anchor gate decides how far the prediction may move from the frozen estimate. The final point map is therefore not an unconstrained average of arbitrary map values; it is the model-space mean of a small residual mixture whose components remain tied to the prior.

The final prior-anchored model should not be read as only the final fine-tuning loss. It has two active training phases: a large-capacity residual/GMM stage with probabilistic losses enabled, followed by an RMSE/MAE-dominant fine-tuning

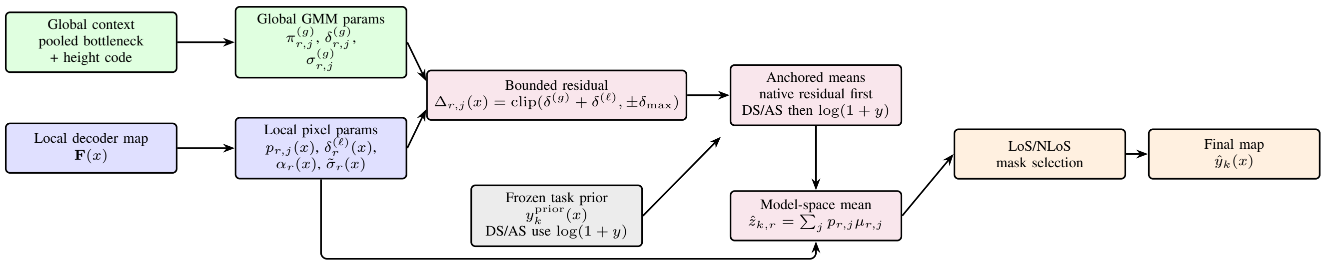

Fig. 5. Final prior-anchored GMM head. The global branch predicts the sample-level residual mixture; the local branch places that distribution pixel by pixel through component weights, residual refinements, gates and local uncertainty. Component means are anchored to the frozen prior and the deterministic map is obtained from the pixel-wise model-space mixture mean after LoS/NLoS selection.

stage. The composite objective implemented by the model is

$$\mathcal{L} = \sum_k w_k\big(\lambda_{\mathrm{NLL}}\mathcal{L}_{\mathrm{NLL},k} + \lambda_{\mathrm{KL}}\mathcal{L}_{\mathrm{KL},k} + \lambda_{\mathrm{mom}}\mathcal{L}_{\mathrm{mom},k} + \lambda_{\mathrm{anchor}}\mathcal{L}_{\mathrm{anchor},k} + \lambda_{\mathrm{guard}}\mathcal{L}_{\mathrm{guard},k} + \lambda_{\mathrm{out}}\mathcal{L}_{\mathrm{out},k} + \lambda_{\mathrm{RMSE}}\mathcal{L}_{\mathrm{RMSE},k} + \lambda_{\mathrm{MAE}}\mathcal{L}_{\mathrm{MAE},k}\big). \tag{26}$$

The probabilistic terms matter during the residual/GMM stage. The NLL term teaches the head to place mixture mass around the transformed target, the KL and moment terms keep the mixture distribution from becoming a numerically convenient but physically uninformative density, and the outlier-budget term discourages a solution that spends most of its capacity on the easy bulk while leaving large tails unconstrained. The final fine-tuning stage then disables those terms and optimizes the deterministic mixture-mean point map for RMSE/MAE, with a small anchor and guard that keep the correction close to the prior. Table VII summarizes the roles. The selected checkpoint is therefore reported as a deterministic RMSE improvement over the frozen priors, not as a calibrated probabilistic predictor. In this final stage, checkpoint selection uses the validation composite $\mathrm{PL}_{\mathrm{RMSE}} + 0.30\,\mathrm{DS}_{\mathrm{RMSE}} + 0.25\,\mathrm{AS}_{\mathrm{RMSE}}$, matching the reported `score_weights`.

The practical role of the two stages is simple. The first stage makes the head behave like a residual density model: it learns where residual mass lies and discourages the old single-mode collapse. The final fine-tuning stage keeps the same architecture but shifts the optimization weight toward the deterministic metrics used for the paper tables. The anchor and guard terms remain active at small weight so the fine-tuned model cannot gain average RMSE by damaging large regions where the frozen prior was already accurate.

V. Experimental Results

A. Main Test Results

Table X reports the final unseen-city test metrics. The model reduces RMSE relative to the frozen prior for all three targets. Path loss is the cleanest success case because both RMSE and MAE improve. Delay spread improves RMSE while MAE slightly increases; this is consistent with reducing rare high-delay errors while introducing small local shifts. Angular spread improves RMSE strongly and is approximately neutral in MAE.

The original numerical targets were 5 dB, 50 ns, and 20°. The final model clears all three on unseen cities. More importantly, it does so jointly: the same network predicts PL/DS/AS instead of using separate specialist models.

B. LoS/NLoS and Split Diagnostics

Table XI separates valid pixels by the geometric visibility mask. The residual model improves the prior in both LoS and NLoS pixels for all targets. This is important because global RMSE could otherwise hide a failure in the physically hard NLoS subset.

The validation/test sanity check is also useful. The validation row reports an external re-evaluation of the selected checkpoint, while the test row is the real generalization result. Delay spread worsens more on test because the held-out city mix contains harder high-delay cases. The frozen delay-spread prior shows the same validation-to-test worsening, confirming that this is a city-mix effect rather than an evaluator change.

C. Prior and Split Breakdowns

The frozen priors are strong enough to deserve their own reporting. They are not simply baselines for the final neural model; they encode most of the stable geometry and therefore determine what residual problem the network sees. Because these prior calibrations are deterministic, non-DL estimators, their standalone audits are reported separately from neural checkpoint selection. The path-loss table uses the held-out validation+test split after fitting the 10840 training references and evaluating 5340 held-out references (4994 topology-assigned rows). The spread table uses the final 14-city test split because Try 79 shares the 10840/2750/2590 train/validation/test contract used by the final Try 80 model. Table XIII shows the height dependence of the calibrated path-loss prior. This is not post-hoc stratification: UAV height is part of the prior estimator through the centred geometry, elevation angle, and height-binned calibration, and it later conditions the residual neural model through FiLM. The trend is strongest in NLoS: the low-altitude bin remains the hardest because many receivers are blocked by local urban structure, while high UAVs see fewer deep shadow pockets. Table XIV shows the matching spread-prior pattern. Delay spread in particular collapses with altitude, from 39.90 ns overall in the lowest bin to 3.70 ns above 300 m. Combined with the empirical HDF5 histogram in Fig. 1, this means the high-altitude rows are not

TABLE VII
LOSS TERMS AVAILABLE IN THE FINAL PRIOR-ANCHORED OBJECTIVE AND THEIR PRACTICAL PURPOSE.

Term	Active stage	Role
GMM NLL	residual/GMM stage	Fits the per-pixel mixture density in native PL space and log-transformed spread space.
Distribution KL	residual/GMM stage	Regularizes the learned mixture so it remains close to the intended distributional structure instead of using degenerate weights or variances.
Moment matching	residual/GMM stage	Encourages the model-space mean and spread of the mixture to agree with target moments, stabilizing deterministic extraction from the GMM.
Outlier budget	residual/GMM stage	Keeps rare large residuals visible in the objective instead of letting the global average hide them.
Anchor	both stages	Penalizes unnecessary deviation from the frozen prior; this is especially important because the prior is already a strong calibrated predictor.
Prior guard	both stages	Adds a penalty when the model becomes worse than the prior at valid pixels, discouraging destructive corrections.
RMSE / MAE	both stages, dominant in final fine-tuning	Optimizes the deterministic map reported in the final evaluation; RMSE handles tails and MAE tracks typical absolute error.

TABLE VIII
TRAINING CONFIGURATION OF THE FINAL LINE.

Item	Value
Image size	513×513
Input channels	9 map channels plus height FiLM conditioning
Backbone	GroupNorm U-Net, base width 128, FiLM dim. 160
Residual head	3 mixture components, bounded residuals
Optimizer	AdamW, learning rate 5×10^{-5}, weight decay 0.01
Batching	batch size 1, gradient accumulation 8
Loss schedule	Large-capacity residual/GMM stage plus final residual fine-tuning; see Table IX

TABLE IX
FINAL PRIOR-ANCHORED LOSS SCHEDULE. THE NON-FINE-TUNED RESIDUAL/GMM STAGE IS PART OF THE FINAL PIPELINE, EVEN THOUGH THE REPORTED CHECKPOINT IS SELECTED AFTER THE DETERMINISTIC FINE-TUNING STAGE.

Setting	Residual/GMM stage	Final fine-tuning
NLL / KL / moment / outlier	1.00 / 0.25 / 0.10 / 0.02	0 / 0 / 0 / 0
RMSE / MAE	1.50 / 0.10	1.00 / 0.05
Anchor / guard	0.05 / 0.10	0.01 / 0.02
Task weights	PL 1.0, DS 1.0, AS 1.0	PL 1.0, DS 0.25, AS 0.25
Learning rate	2×10^{-4}	5×10^{-5}

better because they have more training support; the training split contains only 866 samples above 300 m. They are better despite that scarcity, which indicates a physically easier regime with fewer blocked links and weaker spread tails.

The path-loss prior also varies by topology. Its overall RMSE ranges from 1.43 dB in open sparse low-rise maps to 2.97 dB in dense block high-rise maps. The NLoS branch is hardest in sparse vertical and dense high-rise regimes, reaching 3.91 dB and 4.09 dB; the overall pixel-weighted prior remains 1.92 dB.

The final residual model improves the priors consistently across data splits, not only on the validation split used for model selection. Table XV reports the model-minus-prior RMSE deltas. The test row is the main generalization claim. The all-city row is not used for model selection, but it checks that the gain is not a held-out split artifact.

The all-city morphology/height audit groups samples by broad urban form (`open`, `mixed`, `dense`) and antenna-height bin (`low`, `mid`, `high`). These rows are not nine separately trained final neural networks. They are evaluation buckets that mirror the regime structure used by the frozen priors and fallback tables: morphology is derived from building-density/height thresholds, and the height tag comes from the UAV altitude range. This is a stricter check than the global row because it asks whether the same residual model helps across the deployed prior regimes rather than only in one easy subset. Table XVI reports RMSE deltas for the nine groups. All entries are negative, so the residual correction improves every morphology and height group for all three targets.

D. Why Distribution Diagnostics Matter

Fig. 6 shows the diagnostic that changed the modeling strategy: LoS and NLoS path loss do not form a single easy Gaussian map target. The distribution-first predecessor modeled the marginal with GMM heads and showed that the NLoS collapse was not inevitable. The final model keeps this lesson but anchors the residual around explicit priors.

The internal diagnostic numbers are kept as scale references, not as final claims. The earlier distribution-first path-loss GMM fits reached 4.43 dB LoS and 3.34 dB NLoS pixel-aggregate RMSE, showing that NLoS collapse was a modeling choice rather than a necessity. The earlier distribution-first spread fits reached 30.96 ns delay-spread RMSE and 11.78° angular-spread RMSE, motivating log-domain and mixture-aware treatment of spread targets. The frozen priors then provided stronger anchors: the standalone path-loss prior reached 1.9246 dB RMSE on its held-out validation+test split and 1.9383 dB when frozen on the final test split, while the spread priors reached 28.10 ns delay-spread RMSE and 13.74° angular-spread RMSE on that same final test split.

For spread targets, quantized labels make visual and RMSE interpretation harder. The released delay and angular maps are integer-valued; smooth predictions can be penalized across staircase-like target boundaries even when the morphology is plausible. This does not invalidate RMSE, but it warns against overinterpreting small local residuals.

E. External Metric Context

Table XVII places the final path-loss result beside the closest available path-loss-side references. The table is deliberately compact and cautious: datasets, height assumptions, target definitions, and splits differ. No scalar indoor or link-level spread predictor is used as a direct dense-map leaderboard.

TABLE X
FINAL MODEL TEST RESULTS ON UNSEEN CITIES. NEGATIVE Δ MEANS THAT THE RESIDUAL MODEL IMPROVES THE FROZEN PRIOR.

Target	Model RMSE	Prior RMSE	ΔRMSE	Model MAE	Prior MAE	ΔMAE	Unit
Path loss	1.6519	1.9383	-0.2865	0.9298	1.2319	-0.3020	dB
Delay spread	26.5570	28.1023	-1.5453	7.9897	7.3033	+0.6864	ns
Angular spread	11.3854	13.7416	-2.3562	5.0088	4.9890	+0.0198	deg

TABLE XI
FINAL TEST RMSE SEPARATED BY LOS AND NLOS PIXELS.

Target	Scope	Model	Prior	Δ
PL	LoS	1.4675	1.7519	-0.2844
PL	NLoS	3.1482	3.5143	-0.3661
DS	LoS	25.4419	27.4851	-2.0432
DS	NLoS	29.2045	29.6157	-0.4112
AS	LoS	12.3510	15.2818	-2.9308
AS	NLoS	8.4424	8.6614	-0.2191

TABLE XII
VALIDATION VERSUS HELD-OUT TEST CHECK FOR THE SAME SELECTED MODEL.

Split	Samples	PL RMSE	DS RMSE	AS RMSE
Validation	2750	1.6157	22.1061	11.3782
Test	2590	1.6519	26.5570	11.3854

TABLE XIII
CALIBRATED PATH-LOSS PRIOR RMSE BY UAV HEIGHT ON THE HELD-OUT VALIDATION+TEST SPLIT.

Height bin	Overall	LoS	NLoS	Samples
12–50 m	2.180	1.797	3.756	1211
50–150 m	1.936	1.785	3.228	2634
150–300 m	1.680	1.666	2.259	786
300–500 m	1.627	1.619	2.255	362

TABLE XIV
CALIBRATED SPREAD-PRIOR RMSE BY UAV HEIGHT ON THE FINAL 14-CITY TEST SPLIT.

Height bin	Metric	Overall	LoS	NLoS
12–50 m	DS	39.90	43.04	35.13
50–150 m	DS	27.17	26.72	28.29
150–300 m	DS	8.70	8.79	8.13
300–500 m	DS	3.70	3.80	2.76
12–50 m	AS	15.69	18.81	9.93
50–150 m	AS	14.19	15.89	8.41
150–300 m	AS	10.39	11.07	4.49
300–500 m	AS	8.85	9.34	2.85

TABLE XV
RMSE IMPROVEMENT OVER FROZEN PRIORS BY SPLIT. NEGATIVE VALUES MEAN THE RESIDUAL MODEL IMPROVES THE PRIOR.

Split	Samples	PL	DS	AS
Train	10840	-0.2895	-2.0157	-2.3578
Validation	2750	-0.2758	-1.3433	-2.4396
Test	2590	-0.2865	-1.5453	-2.3562
All	16180	-0.2867	-1.8406	-2.3710

The fair conclusion is narrow but useful. The final path-loss number is in the same band as strong dense radio-map references. It is numerically below the recent Gao et al. outdoor segmentation results, but that comparison remains

TABLE XVI
ALL-CITY FINAL-MODEL RMSE DELTAS BY MORPHOLOGY/HEIGHT AUDIT GROUP. NEGATIVE VALUES MEAN THE FINAL MODEL IMPROVES THE FROZEN PRIOR.

Group	Samples	PL	DS	AS
`dense/high`	1957	-0.3602	-0.6473	-2.6405
`dense/low`	2018	-0.3359	-1.5779	-2.3994
`dense/mid`	2055	-0.3288	-1.4074	-2.8587
`mixed/high`	2156	-0.2931	-0.6794	-2.3186
`mixed/low`	2060	-0.2745	-2.8379	-2.5845
`mixed/mid`	2094	-0.2781	-2.0522	-2.8200
`open/high`	1272	-0.2721	-0.5428	-1.2874
`open/low`	1269	-0.2363	-3.1170	-2.0271
`open/mid`	1299	-0.2941	-1.5789	-1.7804

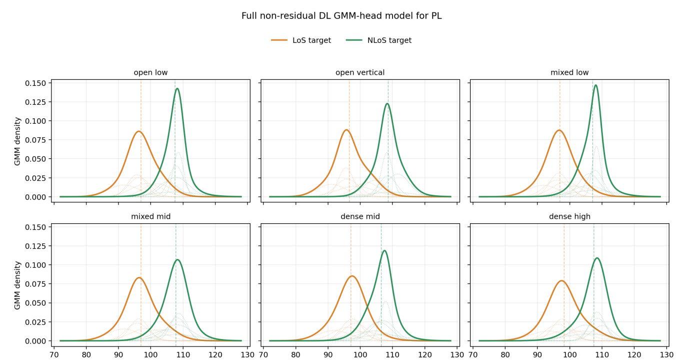

Fig. 6. Full non-residual DL GMM-head model for path loss (PL): target GMM-5 fits by topology group. Thick curves show the summed LoS/NLoS mixture densities, faint dotted curves show the individual Gaussian components, and dashed vertical lines mark the marginal means. The diagnostic exposed different LoS/NLoS marginals and motivated distribution-aware heads before returning to prior-anchored residual learning.

cross-dataset scale context rather than a leaderboard result. The spread results should be interpreted primarily against same-protocol internal baselines because no external dense CKM PL/DS/AS benchmark was found.

F. Qualitative Residual Behavior

Fig. 7 is the qualitative check. The model receives the frozen prior and predicts a correction. The useful visual evidence is not merely that the prediction resembles the target; it is that the correction row tends to anti-correlate with the prior-error row, which is the expected behavior of a prior-anchored residual corrector.

G. Runtime

The final generator was compared against the supplied MATLAB ray-tracing path on Barcelona layouts at full 513×513 resolution on the same laptop: an AMD Ryzen 5 5600H

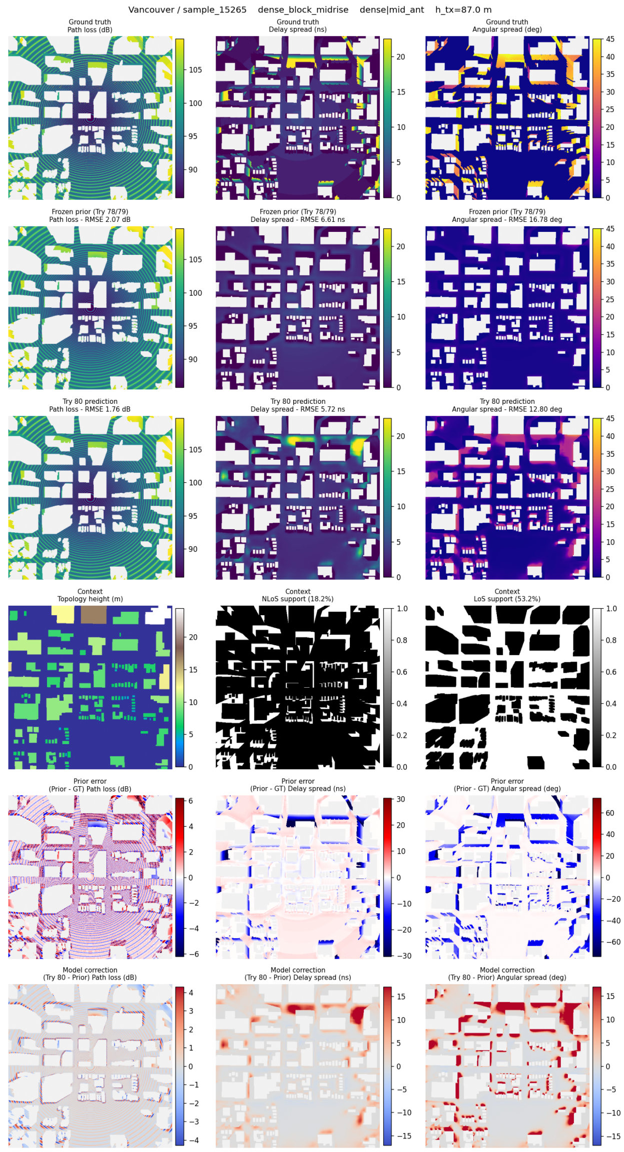

Fig. 7. Representative final-model inspection panel for a dense, mid-antenna held-out test sample in Vancouver. Columns are path loss, delay spread, and angular spread. Rows show ground truth, frozen prior, final prediction, context/support, prior error, and model correction.

CPU and an NVIDIA RTX 3050 Ti Laptop GPU. The surrogate includes LoS/NLoS mask generation, frozen priors, final neural prediction, and default numeric export after the checkpoint is loaded once. It runs in 0.88 s per sample versus 99.89 s for the local ray-tracing path, a $113.7\times$ speed-up under this measurement setup. The ray-tracing row uses the CPU path; the surrogate row uses the laptop GPU with mixed precision.

VI. Discussion

The final result is best interpreted as evidence for a hybrid CKM design rather than as a claim that one backbone solves radio mapping in general. The strongest technical signal is that the residual model improves a high-quality frozen prior across train, validation, test, and every morphology/height audit group. That makes the gain harder to dismiss as overfitting to a single easy city mix. At the same time, the modest size of the delay-spread gain is a useful warning: once the calibrated prior has absorbed the large geometry trend, the remaining spread error is dominated by rare tails and target quantization, not by a smooth missing term. The residual errors that remain are most likely where low UAV elevation, dense local clutter, and NLoS support coincide, because those pixels depend on the particular blocked/scattered path selected by the ray tracer rather than only on broad radial or morphology statistics.

Several limitations remain. First, the labels come from ray tracing rather than field measurements, so the result is simulator agreement, not over-the-air validation. Second, target quantization limits how much sub-unit error structure can be learned. This is not a limitation of the output layer: the model outputs continuous PL/DS/AS values. It is a limitation of the stored labels. For path loss, a 0.9298 dB final MAE against labels quantized at about 1 dB is already close to the label step, even though the RMSE still reveals harder tails. Third,

TABLE XVII
COMPACT COMPARISON WITH NEARBY PATH-LOSS-SIDE METRICS.

Reference	PL-side metric	Reading / caveat
RadioUNet [5]	1.6 dB–3.1 dB equivalent PL RMSE	Final PL is in this band; no UAV height or spread targets.
RadioGUNet [10]	1.304 dB DPM; 1.936 dB IRT	Final PL is between those values; outdoor RadioMapSeer setting.
PMNet / ICASSP [7], [8]	approx. 1.38 dB converted score	Final PL is close but higher under a different benchmark contract; scale context only, no joint spread prediction.
Gao et al. [12]	5.59 dB ICASSP 2023; 12.19 dB ITU 2024	Final PL is numerically lower, but this is scale context rather than a like-for-like win: recent corridor-weighted outdoor PL predictor, no variable-height UAV or joint spread outputs.
ReVeal [15]	1.95 dB outdoor sparse-measurement RMSE	Final PL is in the same range; sparse mapping rather than dense generation.
RMTransformer / PathFinder [9], [11]	$\approx$ 1.82 dB / $\approx$ 1.19 dB converted references	Useful scale checks, but random/specialized splits and different assumptions.
AIRMap [13]	Path-gain error $< 4\,\mathrm{dB}$ against ray tracing	Same broad digital-twin motivation, but path gain and a fixed 200×200 input grid with variable physical extent.
Final model (ours)	**1.6519 dB PL; 26.5570 ns DS;** $11.3854°$ **AS**	**Strict CKM city holdout, centred variable-height UAV, dense** 513×513 **PL/DS/AS.**

TABLE XVIII
RUNTIME COMPARISON FOR ONE FULL-RESOLUTION BARCELONA CKM SAMPLE ON THE SAME RYZEN 5 5600H / RTX 3050 TI LAPTOP HARDWARE.

Method	Timed path	Time
MATLAB RT	Pixel receivers outside buildings; PL/DS/AS parsed from paths; CPU path	99.89 s
Final generator	Masks, priors, PL/DS/AS prediction, numeric export; CUDA mixed precision	0.88 s

the final model is selected under one city-holdout split and one final training seed. Fourth, the dataset fixes frequency, antenna pattern, and simulator assumptions; deployment would require conditioning or calibration across those variables.

VII. CODE AVAILABILITY

The reproducibility record is public as live source repositories. The citations below identify the public source locations and snapshot commits or tags used for this manuscript. The standalone CKM Generator repository contains the Streamlit interface, command-line generator, frozen calibration files, final-model runtime code, and the published checkpoint path used by the demonstrator [31]; it is the inference/demo entry point. The final training and evaluation implementation is collected separately in the final-code repository [32]; it is the source record for reproducing the paper metrics. For traceability, the broader development history and the LaTeX thesis source are also archived in public repositories [33], [34]. Large data and model artifacts are treated as external artifacts or Git LFS objects rather than ordinary source files.

VIII. CONCLUSION

This paper presents a dense variable-height UAV CKM surrogate that jointly predicts path loss, delay spread, and angular spread on unseen cities. Under the strict city-holdout contract, the final model reaches 1.6519 dB, 26.5570 ns, and $11.3854°$, respectively, while using one centred UAV generator across the sampled height range. The result is not only a neural architecture result. Its main contribution is a hybrid recipe: first build strong train-only calibrated priors for the structure that is predictable, then learn bounded residual distributions around those anchors.

The calibrated priors are themselves an important part of the method. They put together coherent two-ray LoS path loss, height-dependent calibration, morphology-aware NLoS correction, visibility switching, and log-domain DS/AS spread statistics. These frozen anchors already explain most radial, visibility, height, and spread-scale structure; the final residual model then improves them by 0.2865 dB, 1.5453 ns, and $2.3562°$ RMSE on the held-out test set. Continuous UAV height is handled through both the height-aware priors and a sinusoidal FiLM conditioning path, so the same model covers the varying-height regime without per-altitude retraining.

The distribution-first diagnosis remains the key modelling lesson. NLoS path-loss residuals are mixture-like and prone to mode collapse, while DS/AS targets are quantized and spike-plus-tail rather than single smooth residuals. GMM-style residual heads and histogram checks therefore matter because they prevent visually plausible maps from hiding collapsed marginals. The remaining spread errors are consistent with this interpretation: the large-scale DS/AS structure is usually recovered, but rare high-spread pixel groups can still be missed, and some false positive high-spread patches remain.

The main practical conclusion is narrow and useful. For this fixed ray-traced CKM simulation contract, the prior-anchored residual GMM-head model is accurate, fast, and interpretable enough to replace repeated map generation for many planning experiments, achieving a measured $113.7\times$ speed-up over the local MATLAB ray-tracing path. The result is a reproducible surrogate for a defined dataset, split, carrier, antenna setup, and receiver mask, and its clearest lesson is methodological: diagnose the target distribution, freeze the stable structure in auditable priors, and spend neural capacity on the residual structure that the priors cannot explain.

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